SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 6

Marketing Analytics

Measurement foundations, customer analytics (STP, RFM, CLV, survival, clustering, HHI), promotion and pricing (causal impact, elasticity, optimal price, market basket, forecasting, game theory), Marketing Mix Modeling (adstock, ROI, budget optimisation, Monte Carlo, MILP), location analytics and the analysis-to-decision framework. Cases: Eiger vs Torch, Indonesian ground coffee.

Material

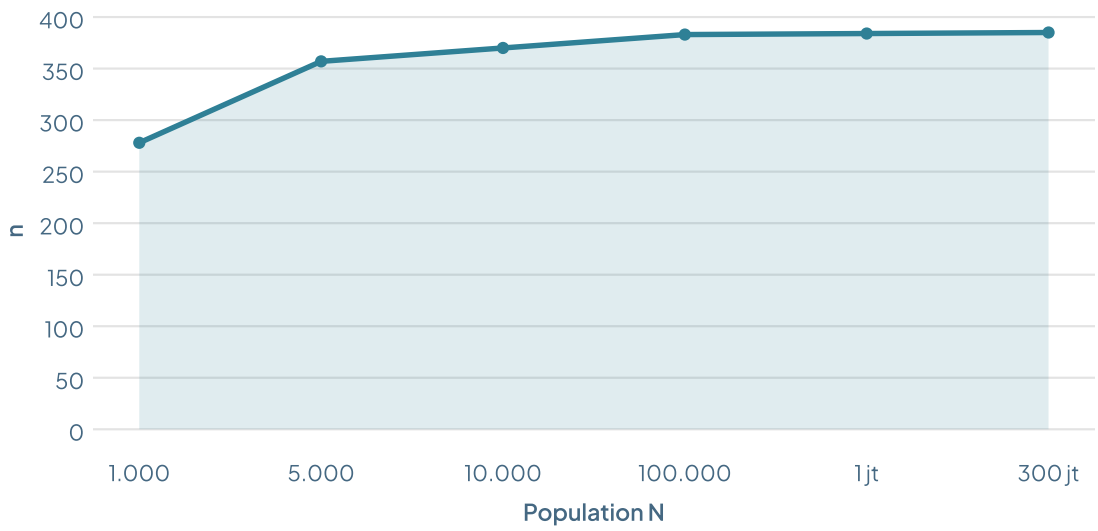
1 Framing and measurement foundations

Research vs analytics vs BI; three questions and three layers; KPI trees; data quality and seven biases; sampling vs non-sampling error; Cochran sample size (~385 for ±5%); survey-cheater detection (rules vs Isolation Forest); Jabodetabek is a sophisticated microcosm, not a miniature; hierarchical Bayes shrinks small samples; Simpson's paradox.

2 Sample size: why ~385 works for a city or a country

Deep dive

Sample size vs population (±5%, 95%)



$n_0 \approx 385$ for ±5% at 95%

Beyond tens of thousands, population size barely matters.

3 Customer analytics

STP and segment criteria; clustering (k-means, k-prototypes, elbow, LPA). RFM map of actions. CLV = $\Sigma\, m \cdot r^t/(1+d)^t$; CAC < CLV per segment. Survival: discount joiners churn by month 4.7 vs 8.2 (HR 0.585). HHI: telco 3,177, cars 1,884. Pair attitude metrics with behaviour; uplift modelling.

4 RFM: from scores to actions

Deep dive

↕ Top row: high frequency & monetary · bottom: low

Champions
Reward, ask referrals.

Can't lose them
Win back now.

New / Promising
Onboard.

Hibernating
Cheap reminders.

↔ Left column: recent buyers · right: lapsed

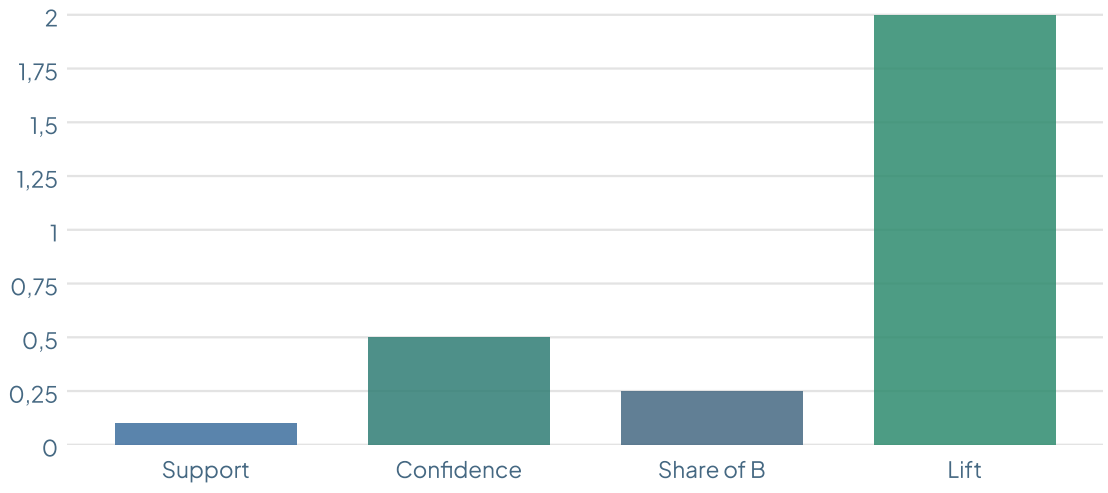
5 Promotion and pricing

Causal impact: +IDR 5.3m/day (+4.2%) vs a synthetic control. Elasticity: +15% price, –10% volume → +3.5% revenue. QTY = 200 – 2P with cost 20: revenue peak at 50, profit peak at 60. Apriori: support, confidence, lift (>1 bundle). Forecasts need ranges (Red River 1997). Discounts are strategic games.

6 Optimal price, elasticity and market basket Deep dive

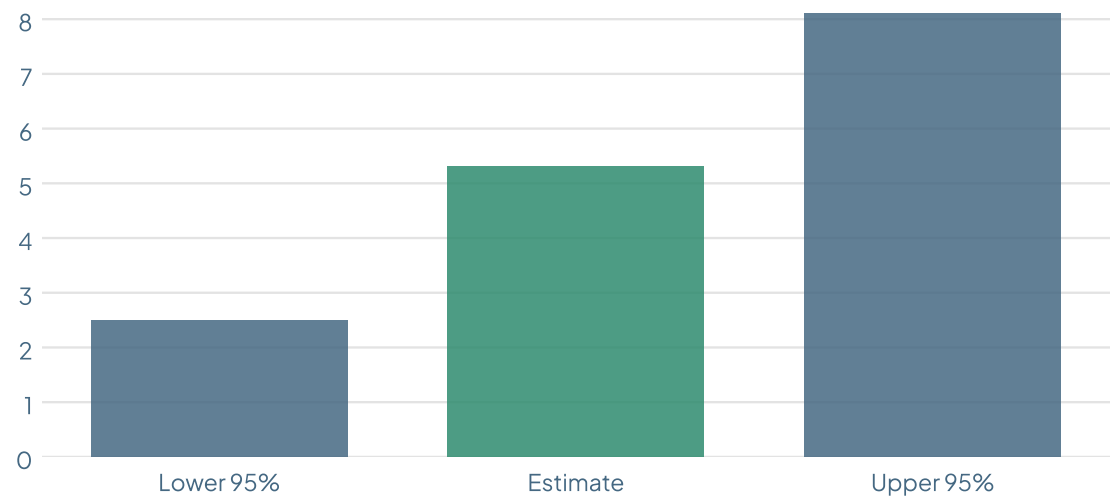
Revenue peaks at 50k, profit at 60k.

Nutrimart basket metrics



Lift 2.0 > 1 → complements.

Flash-sale causal impact per day (IDR m)



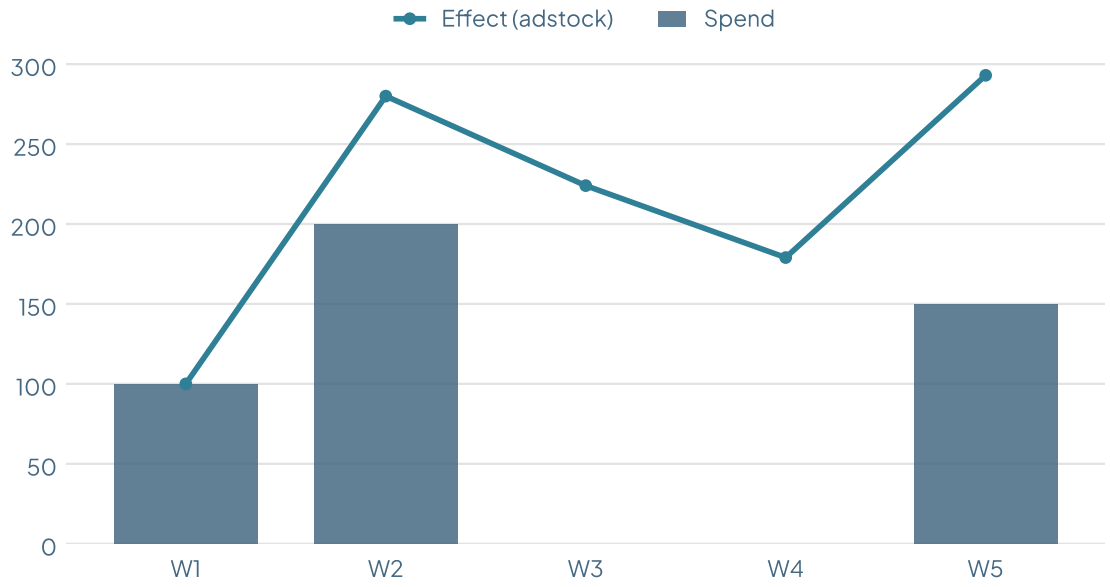
+4.2%, cumulative +159m.

7 Marketing Mix Modeling

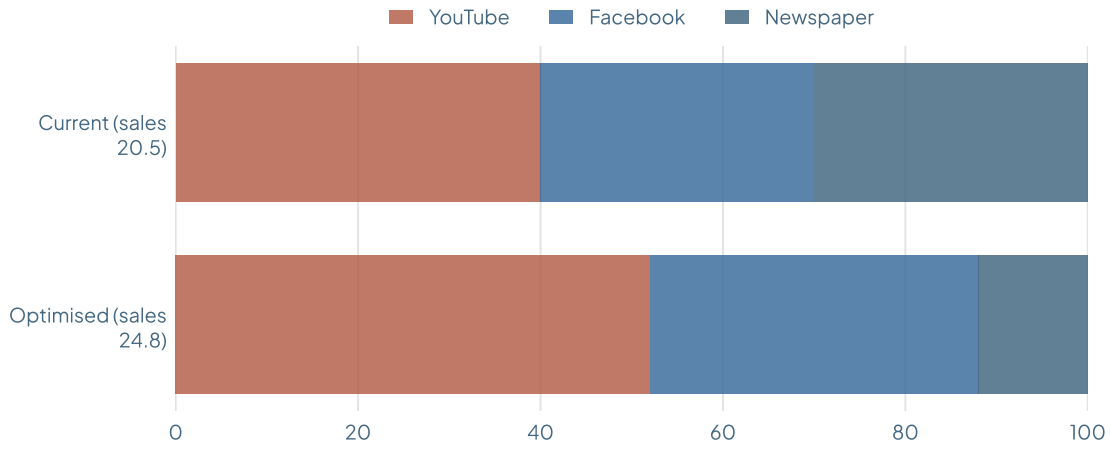
MMM regresses all-channel spend on sales. Adstock with decay 0.8 (100, 280, 224, 179, 293). Example: YouTube +6.2, Facebook +3.1, newspaper −0.8; reallocating at fixed budget lifts sales 20.5 → 24.8. Monte Carlo sweet spot IDR 275m. MILP with fixed costs: email only above IDR 8m; IDR 34m vs 30m. Nowcasting: tight ≠ right.

8 MMM in charts: adstock and reallocation Deep dive

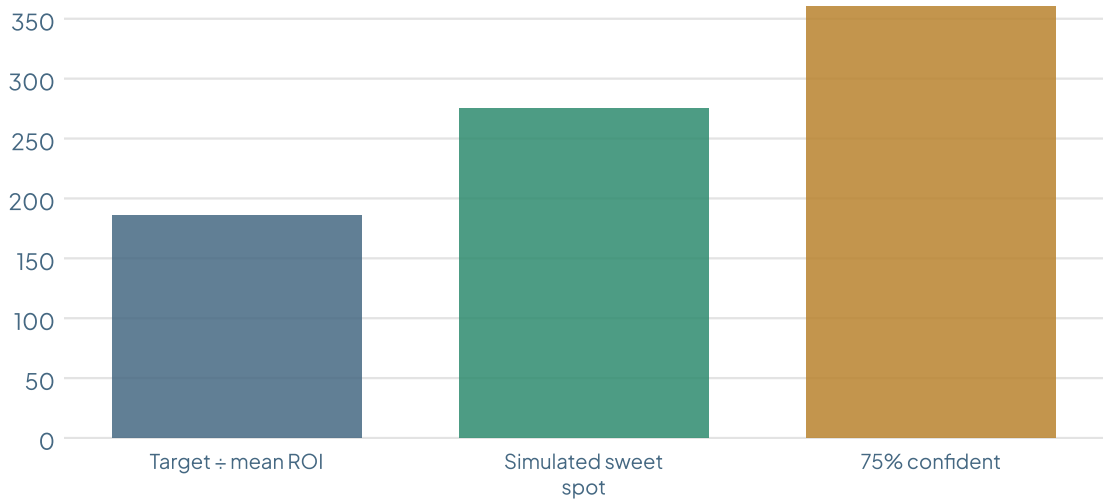
Adstock (decay 0.8): spend vs effect



Budget split: current vs MMM optimiser



Budget: average maths vs simulation



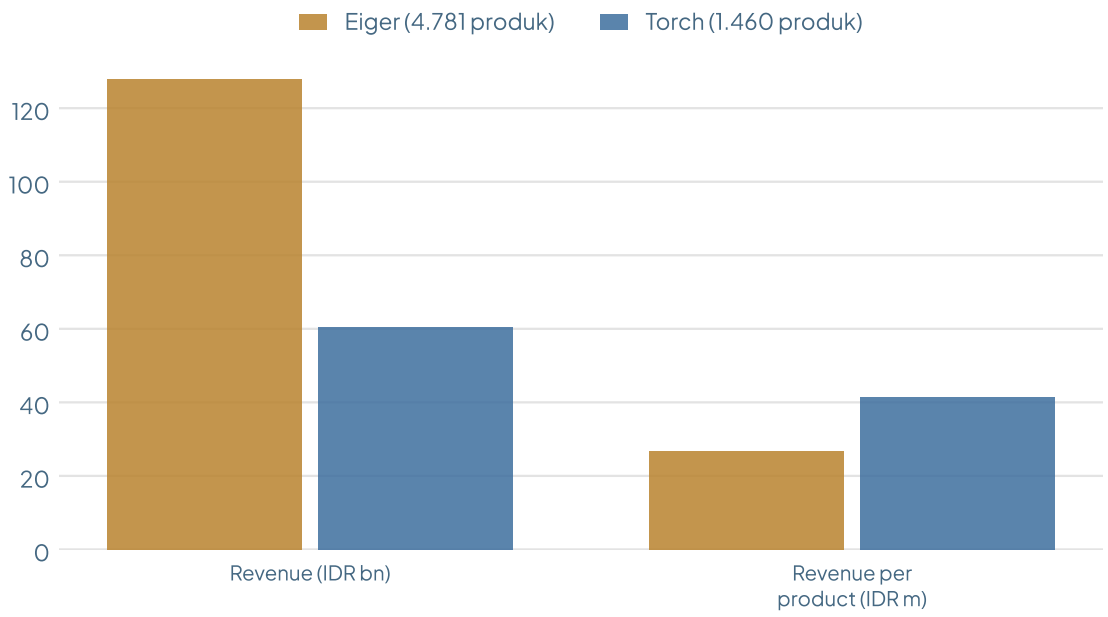
Simulation: IDR 275m is the sweet spot (67.8% chance of profit).

9 Location analytics and analysis to decision

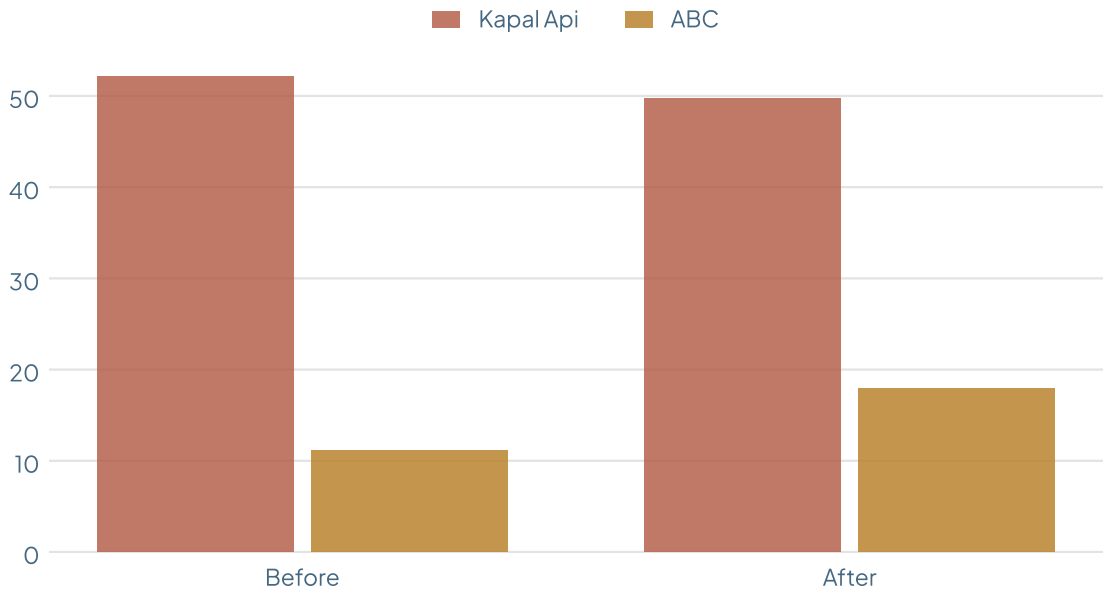
Voronoi maps (Bekasi padel and laundries; overlay population). The five-step framework: question with deadline, one primary metric, data sources, method and confidence, decision rule with guardrail. GenAI speeds work but must never be the source of numbers. Seven distinctive brand assets.

10 Eiger vs Torch and the coffee war Deep dive

Eiger vs Torch on Tokopedia (May 2025)



Kapal Api vs ABC share (%)



Share lost by Kapal Api moved to ABC, still inside Santos Jaya Abadi.

Key takeaways

1. Decision, data, confidence.
2. Design before sample size.
3. RFM, CLV, survival.
4. Causal impact needs a control.
5. Optimise contribution, not revenue.
6. MMM with adstock.
7. AI is not a source of numbers.

Case studies

Case & worked answers

Eiger vs Torch on Tokopedia; the ground coffee war

Eiger goes wide (4,781 products, IDR 128bn, 26.8m per product); Torch goes deep (1,460 products, IDR 60.6bn, 41.5m per product, mostly bags). Analyse margin per category/SKU and Pareto before choosing. Coffee: Santos's multi-brand encirclement holds ~60%; Top Coffee's huge ad spend bought ~3% share; Luwak White won by creating a category.

Worked examples

Problem 1 · Easy · **Elastisitas harga**

Harga naik dari Rp10.000 ke Rp11.000, volume turun dari 1.000 ke 880 unit. Hitung elastisitas dan perubahan pendapatan.

1. %Δ harga = +10%; %Δ volume = −12%.
2. Elastisitas = 12% ÷ 10% = 1,2 → elastis (> 1).
3. Pendapatan: 10 juta → 11.000 × 880 = 9,68 juta (−3,2%).

Answer: e = 1,2 (elastis); pendapatan turun 3,2%.

Problem 2 · Medium · **Harga optimum**

Permintaan Q = 500 − 10P (P dalam ribu rupiah), biaya variabel Rp10 ribu. Cari harga yang memaksimalkan omzet dan yang memaksimalkan laba kontribusi.

1. Omzet maksimum di P = a ÷ 2b = 500 ÷ 20 = 25.
2. Laba maksimum di P = (a/b + VC) ÷ 2 = (50 + 10) ÷ 2 = 30.
3. Di P = 30: Q = 200, laba = (30 − 10) × 200 = 4.000 (ribu).
4. Di P = 25: Q = 250, laba = 15 × 250 = 3.750 → lebih kecil.

Answer: Omzet maksimum di Rp25 ribu, laba maksimum di Rp30 ribu (Rp4 juta).

Problem 3 Easy **Market basket**

Dari 2.000 transaksi: roti 400, selai 250, roti + selai 150. Hitung support, confidence (roti → selai), dan lift.

1. Support = 150 ÷ 2.000 = 7,5%.
2. Confidence = 150 ÷ 400 = 37,5%.
3. Porsi selai sendiri = 250 ÷ 2.000 = 12,5%.
4. Lift = 37,5% ÷ 12,5% = 3,0.

Answer: Lift 3,0 > 1 → komplemen kuat; cocok dibundel atau diletakkan berdekatan.

Problem 4 Medium **Adstock dan ukuran sampel**

(a) Belanja iklan 3 minggu: 100, 0, 50; decay 0,5. Hitung efek adstock. (b) Berapa responden untuk ±4% (95%) di populasi 2.000?

1. (a) Minggu 1: 100. Minggu 2: 0 + 0,5 × 100 = 50. Minggu 3: 50 + 0,5 × 50 = 75.
2. (b) $n_0 = 1{,}96^2 \times 0{,}25 \div 0{,}04^2 = 600{,}25$.
3. Koreksi populasi: $n = 600{,}25 \div [1 + 599{,}25 \div 2.000] = 600{,}25 \div 1{,}2996 = 461{,}9 \rightarrow 462$.

Answer: (a) 100, 50, 75. (b) 462 responden.

Problem 5 Medium **Customer lifetime value (CLV)**

A customer brings IDR 600 thousand of margin a year. The yearly retention rate is 80% and the discount rate 10%. Compute CLV over a 3-year horizon.

1. $CLV = \Sigma\ m \cdot r^t \div (1 + d)^t$
2. t = 1: 600 × 0.8 ÷ 1.1 = 436.4. t = 2: 600 × 0.64 ÷ 1.21 = 317.4. t = 3: 600 × 0.512 ÷ 1.331 = 230.8.
3. CLV = 436.4 + 317.4 + 230.8 ≈ **IDR 984 thousand**.
4. So acquisition costs above about IDR 984 thousand will not be recovered in 3 years. Raising retention from 80% to 85% lifts CLV noticeably — retention is often cheaper than acquisition.

Answer: CLV ≈ IDR 984 thousand.

Practice questions

Try answering first. The answer key is below each question.

1. Lift < 1 means…

A. Complements
B. Substitutes, don't bundle
C. No data
D. Equals support

Answer: B. Bought together less than chance.

2. Q = 200 − 2P, VC 20. Profit-max price…

A. 40
B. 50
C. 60
D. 100

Answer: C. (100 + 20)/2.

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3. Decay 0.8, spend 200, prior effect 100. Week-2 effect…

A. 200
B. 260
C. 280
D. 300

Answer: C. 200 + 80.

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4. Frequent buyer, silent 7 months, is…

A. Champions
B. Can't lose them
C. New
D. Promising

Answer: B. Win back now.

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5. 4× respondents makes sampling error…

A. Same
B. Half
C. Quarter
D. Zero

Answer: B. Falls with $\sqrt{n}$.

Sources

- Marketing Analytics — MK6 Week 6
- Chapman & Feit (2019), R for Marketing Research and Analytics
- Kotler & Keller